

A Novel Technique for the Simultaneous Determination of ^{210}Pb and ^{210}Po Using a Crown Ether

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ABSTRACT

A simple radiochemical procedure has been developed to separate lead and polonium simultaneously from different environmental and biological materials. After adding ^{208}Po tracer and lead carrier samples are decomposed using mineral acids. Lead and polonium are selectively retained from 2 M HCl solution by the bis-4,4'(5')-t-butyl-cyclohexano-18-crown-6-ether; polonium is stripped with 6 M HNO_3 while lead is stripped with 6 M HCl solution. A polonium source is prepared by spontaneous deposition onto a silver disk. The activity concentration of ^{210}Po is determined by isotope dilution alpha spectrometry. Lead is precipitated as oxalate; the chemical recovery is determined by gravimetry. The activity concentration of ^{210}Pb is calculated from the liquid scintillation spectrum. © 1997 Published by Elsevier Science Ltd

INTRODUCTION

Lead-210 and ^{210}Po are relatively long-lived members of the natural ^{238}U decay series. They contribute significantly to the radiation dose of the population according to the UNSCEAR Report from 1988, 8% of the

natural internal radiation dose to man originates from these two nuclides. Because of their longer half-lives (22 years and 138 days, respectively) and chemical differences compared to other chain members, these two radio-nuclides are expected not to attain radioactive equilibrium with the ^{238}U parent nuclide or other precursors of the series as the long-lived ^{234}U , ^{230}Th or ^{226}Ra in the environment except for specially stable, immobility favouring conditions that exist in rocks and ores. In addition to natural tendencies towards separation of parent and daughter nuclides, industrial activities can have a major impact on enhancement of the radioactivities of certain nuclides, such as production of phosphate fertilizers, mining and processing of uranium etc. (Steinhauser, 1990). Certain biological organisms (bioindicators) are known to be able to accumulate various elements, e.g. marine mussels take up polonium preferentially (Yamamoto *et al.*, 1994). Because of the lack of radioactive equilibrium in biological and most of the environmental samples, the individual analysis of both ^{210}Pb and ^{210}Po is desirable. If there is a complete secular equilibrium in a sample, ^{210}Pb and ^{210}Po results can be regarded as parallel ones contributing to quality control.

Lead-210 decays by the combined emission of weak beta and gamma rays and internal conversion electrons to the ground state of ^{210}Bi , which in turn decays, according to its 5-day-long half-life, by the emission of beta particles to ^{210}Po . ^{210}Po is a pure alpha emitter. Neither ^{210}Pb nor ^{210}Po can be analyzed by common instrumental measuring techniques because of the significant self-absorption of the weak beta particles (beta-max 16 and 63 keV), conversion electrons (8, 30 and 43 keV), gamma radiation (46 keV) of the former and the alpha radiation of the latter nuclide. This is why lead and polonium are frequently separated chemically prior to their individual analyses.

A great variety of methods have been proposed and applied to separate polonium and lead both individually and simultaneously. Polonium is separated most frequently by spontaneous or electrochemical deposition onto metal discs, most probably silver. Chemical concentration prior to deposition is not always performed (RADREM, 1989), and the presence of interfering ions makes the procedure less reliable both by reducing the deposition yield of polonium and increasing the thickness of the deposited layer. Thus the alpha spectra can be degraded, in extreme cases even preventing the alpha spectrometric determination of the polonium nuclides. The different concentration methods are based mainly on solvent extraction (Barnabee & Sill, 1981); anion exchange (Church *et al.*, 1994; Moser, 1993), coprecipitation (USEAC, 1972; USEPA, 1979), or distillation (RADREM, 1989) processes. On the other hand there was no highly selective method for the separation of lead until recently. This was revealed by the great variety of methods that included solvent extraction (USEAC, 1972), ion exchange

(Walton and Rocklin, 1990), extraction chromatography (Sebesta *et al.*, 1975), precipitation (Alfassi and Wai, 1992) and combinations thereof. In 1991, Horwitz *et al.* (Horwitz *et al.*, 1992, 1994) developed a novel extraction chromatographic material (Sr. Spec) for the separation of strontium that also showed extremely high selectivity for lead. This year a new article was published describing the preparation of a modified version of this extraction material (Pb. Spec) with reduced selectivity for lead thus allowing removal from the resin [15]. The specific component of both solid extractants is the same crown ether, i.e. the bis-4,4'(5')-[tert.-butylcyclohexano]-18-crown-6. The Pb. Spec material was applied to the isolation and mass spectrometric determination of lead isotopes in geological samples.

Our idea was to use the unmodified Sr. Spec resin for the simultaneous determination of lead and polonium. Sr. Spec is known to retain polonium from nitric acid solutions. According to Horwitz *et al.* (1992), the maximum distribution coefficient for polonium is about 10^2 in 0.5–1.0 M HNO_3 ; at higher acid concentrations it is reduced abruptly. (The distribution ratio for lead is higher than 10^3 in a wide HNO_3 concentration range from 0.5 M up to 4 M.) Even under the conditions of maximum polonium retention, lead and polonium could not be separated efficiently in our chromatographic test experiments using ^{210}Pb – ^{210}Bi – ^{210}Po tracers, because polonium was released from the column before bismuth had been eluted completely. However simultaneous separation of lead and polonium can be performed from a HCl solution. Lead is retained by Sr. Spec from HCl less strongly than from HNO_3 at the same concentrations, thus allowing its easy removal from the column; also polonium is retained quantitatively. Polonium can be eluted with 6 M HNO_3 , and lead is finally stripped with 6 M HCl.

In this paper the following topics are to be discussed: (i) distribution ratios for lead, bismuth and polonium on Sr. Spec from HCl solutions determined by batch technique; (ii) selectivity data of different elements on Sr. Spec from 2 M HCl solution obtained by column chromatographic technique, tracer studies and ICP-AES; (iii) major parameters of the separation procedure and the detection methods; and (iv) results of analysis of reference materials.

EXPERIMENTAL

Reagents

Sr. Spec (particle size: 100–150 μm) was obtained from Polonium-208 Eichrom Ind. Inc. All chemicals were analytical reagent grade and were

used as received. Polonium-208 tracer was purchased from AEA, Harwell and treated according to the procedure of R. P. Bernabee and C. W. Sill (1981). Lead-210 tracer was obtained from Czech Metrological Institute, Inspectorate for Ionizing Radiation, Prague. Reference materials were supplied by the IAEA, Vienna and the National Institute of Standards and Technology, Gaithersburg.

Methods

Determination of distribution ratios of lead, bismuth and polonium on Sr. Spec in HCL solutions (Fig. 1)

5 ml of HCL solutions of different concentrations were spiked with a tracer containing equilibrium activities of ^{210}Pb , ^{210}Bi and ^{210}Po . 100 mg of Sr. Spec were added to each and shaken for 2 h at room temperature. One-millilitre aliquots of the settled solutions were filtered, analyzed by LSC (see next sub-section) and relative count rates compared to those before separation were determined. Distribution ratios were calculated from the following equation:

$$K_d = \frac{I_0 - I_t V_a}{I_t V_r}$$

where I_0 and I_t are the aqueous phase count rates (cpm) before and after equilibration, respectively, V_r is the volume of the resin (ml) and V_a is that of the aqueous phase (ml).

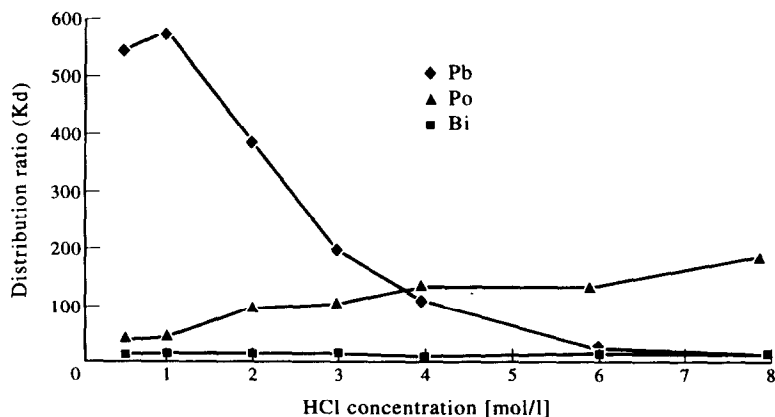


Fig. 1. Distribution ratios K_d of lead, bismuth and polonium in HCL solutions.

Preparation of chromatographic columns

3 g of Sr Spec (100–150 μm) were soaked in distilled water for 1–2 h, then filled into a glass chromatographic column (inner diameter 10 mm) and washed with 100 ml of distilled water, 100 ml 1 M HNO_3 and 100 ml 2 M HCl . It should be noted that columns can be reused several times. To regenerate a used column, 50 ml of distilled water, 100 ml of 0.1 M EDTA (pH adjusted to 7 with NH_3), 50 ml of distilled water and 100 ml of 2 M HCl are passed through it.

Elution behaviour of lead, bismuth and polonium (Fig. 2)

20 ml of 2 M HCl solution spiked with an aliquot of ^{210}Pb tracer in secular equilibrium with ^{210}Bi and ^{210}Po , containing 15 or 30 mg of lead carrier as PbCl_2 were loaded in a Sr. Spec column. The column was then eluted with 100 ml of 2 M HCl , then with 6 M HNO_3 followed by 6 M HCl , 10 ml fractions were collected from the effluent that were gently evaporated with some drops of 65% HNO_3 . Residues were taken with 1 ml 6 M HNO_3 , mixed with 15 ml Insta gel and analyzed by the LSC technique. Net count rates of the nuclides compared to those in the load solution were determined as discussed below.

Elution behaviour of selected ions (Table 1)

One gram of old ash of the standard reference material NBS 4354 (sediment) of known composition was spiked with 8 dpm ^{208}Po tracer. Thirty

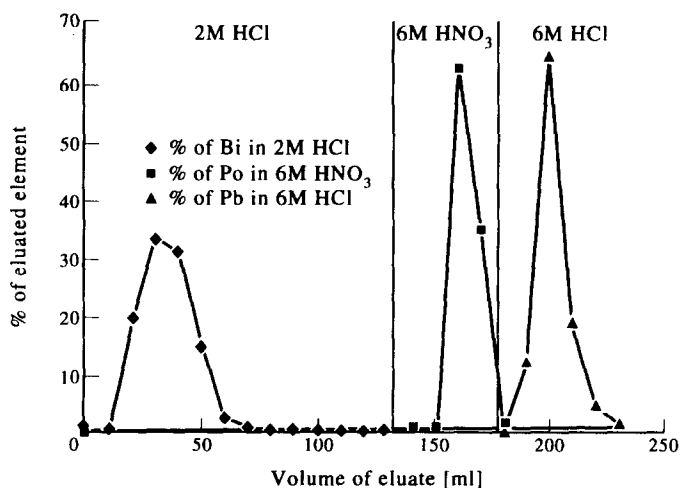


Fig. 2. Elution chromatogram of lead, bismuth and polonium on Sr. Spec.

TABLE 1
 Elution Behaviour of Selected Elements on Sr. Spec Column (% of the Elements in the Given Fractions Determined by ICP-AES
 load: 0.5 g NBS 4354 sediment + 15 mg Pb + 0.07 Bq ^{208}Po)

Element	Effluent 10 ml 2 M HCl	Eluate 25 ml 2 M HCl	Eluate 25 ml 2 M HCl	Eluate 25 ml 2 M HCl	Po strip 25 ml 6 M HNO ₃	Po strip 25 ml 6 M HNO ₃	Pb strip 25 ml 6 M HCl	Pb strip 25 ml 6 M HCl
Pb	0	0	0	0	0	0	2	0
Zn	61	36	0	0	1	1	0	0
P	70	30	0	0	0	0	0	0
Co	75	25	0	0	0	0	0	0
Fe	72	28	0	0	0	0	0	0
Cr	76	24	0	0	0	0	0	0
Mg	74	26	0	0	0	0	0	0
Ca	73	26	0	0	0	1	0	0
Cu	71	24	0	1	0	1	2	0
Ti	76	24	0	0	0	0	0	0
Zr	74	26	0	0	0	0	0	0
Y	75	25	0	0	0	0	0	0
Eu	73	24	0	0	0	0	3	0
Ce	75	25	0	0	0	0	0	0
Al	76	24	0	0	0	0	0	0
Sr	0	99	1	0	0	0	0	0
Ba	0	73	26	0	0	0	0	0

milligrams of lead carrier were added and the material was dissolved with mineral acids (see the destruction procedure in the next paragraph). The sample was taken with 20 ml of 2 M HCl and was divided into two parts: one half was loaded onto a Sr. Spec column, while the other half was analyzed directly as a reference sample. The column was eluted with 4×25 ml 2 M HCl, 3×25 ml 6 M HNO_3 and 3×25 ml 6 M HCl. The composition of the reference sample, and the 25 ml fractions of the effluent were analyzed by ICP-AES. Besides lead, 16 different elements were determined, and their relative concentrations compared to the reference sample were calculated.

Chemical procedure for the separation of lead and polonium from different samples (Table 2)

Samples of 0.5–35.0 g were taken. Thirty milligrams of lead carrier as $\text{Pb}(\text{NO}_3)_2$ and 8 or 16 dpm of ^{208}Po tracer were added. Blank experiments were also performed using 30 mg of Pb and 16 dpm of ^{208}Po .

Destruction procedure for sediment, soil and ore samples. Samples were digested in teflon beakers and evaporated to dryness with the following portions of mineral acids: to 5 g of sediment or soil, $3 \times (10 \text{ ml } 65\% \text{ HNO}_3 + 40 \text{ ml } 40\% \text{ HF})$; $3 \times 30 \text{ ml } 65\% \text{ HNO}_3$; 2.0 g of $\text{H}_3\text{BO}_3 + 30 \text{ ml } 32\% \text{ HCl}$; $2 \times 30 \text{ ml } 32\% \text{ HCl}$ were added. The final residues were taken up with 30–50 ml of 2 M HCl, warmed for about 30 min to promote dissolution, and finally cooled and centrifuged and/or filtered through filter paper. Filtrates were taken for the analysis.

Destruction procedure for biological samples. To 5 g of materials of high organic content (biological samples), 30 ml portions of 65% HNO_3 were added as many times as the organic content seemed to be destroyed. Finally $3 \times 30 \text{ ml}$ of 32% HCl were added and the residue was taken with about 30–50 ml of 2 M HCl. To samples very likely not to contain silica, HF and H_3BO_3 were not given. For different amounts of samples the amounts of chemicals were modified proportionally, but final residues were taken up with 30–50 ml of 2 M HCl.

Chromatographic separation of lead and polonium. The 2 M HCl load solutions were passed through the column, the beakers were rinsed with $2 \times 5 \text{ ml}$ HCl and 90 ml 2 M HCl, and finally 25 ml 6 M HNO_3 were added to remove the non-retained ions from the column. Polonium was stripped with 60 ml 6 M HNO_3 and lead with 60 ml 6 M HCl.

TABLE 2
Separation Scheme of Lead and Polonium Using Sr. Spec

Dry sample weight: 0.1–50 g,
Addition of 30 mg Pb carrier,
Addition of ^{208}Po tracer.

Sample dissolution

Evaporation with mineral acids:
65% HNO_3 , 32% HCl ;
optionally: 40% HF , 32% $\text{HCl} + \text{H}_3\text{BO}_3$
dissolution in 30–50 ml 2 M HCl ;
filtration

Chromatographic separation

on Sr. Spec column

Column preparation: 3 g Sr. Spec;
column conditioning with 100 ml 2 M HCl ;
loading;
elution with 100 ml 2 M HCl , 25 ml 6 M HNO_3 .

Stripping of Po

60 ml 6 M HNO_3

Po source preparation

for alpha spectrometry

evaporation;
dissolution in 0.1 M HCl ;
spontaneous deposition onto Ag disc.

Alpha spectrometry

yield determination based on ^{208}Po

^{210}Po

Stripping of Pb

60 ml 6 M HCl

Pb source preparation

for LSC

evaporation;
evaporation with 3×2 ml 65% HNO_3 ;
Pb oxalate precipitation;
gravimetric yield determination;
sample dissolution with 1 ml 6 M HNO_3 ;
mixing with 15 ml Insta gel.

LSC

^{210}Pb

Preparation of polonium sources. The polonium strip solutions (6 M HNO_3) were carefully evaporated to dryness. Residues were taken with 10 ml of 1 M HCl , the pH of the solution was adjusted to 1 using 10 M NaOH . The solution was transferred to a teflon electrolysis cell containing a polished silver disc of 150 mm^2 active surface at the bottom of the cell.

The beaker was further rinsed with 2×2.5 ml of 0.1 M HCl. Polonium was spontaneously deposited from the stirred solution at about 80°C during 90 min. After completion of deposition, the silver disc was removed, rinsed with distilled water and dried at room temperature.

Preparation of lead sources. The Pb strip solutions were evaporated to dryness. Lead was converted from chloride to nitrate by evaporating it with 3×2 ml of 65% HNO_3 . The residue was taken in 20 ml 1 M HNO_3 , 400 mg of oxalic acid were added to the warm solution and the pH of the solution was adjusted to 6–7 with ammonia. The solution was cooled and filtered through preweighed filter paper (diameter 24 mm), and washed with 3×2 ml distilled water and 2 ml alcohol. Samples were dried below 50°C for half an hour, cooled in a desiccator and weighed to determine the mass of lead oxalate in order to calculate the chemical recovery.

The lead oxalate precipitates, together with the filters, were transferred to liquid scintillation vials; the precipitates were dissolved with 1 ml 6 M HNO_3 and mixed with 15 ml Insta gel. They were kept in the liquid scintillation counter for more than 2 h before counting them, in order to cool the samples, to eliminate fluorescence effects and to permit short lived lead nuclides and their daughter nuclides to decay. Samples were recounted after 2 days.

Counting conditions, instruments

Analysis of ^{210}Po and ^{208}Po by α -spectrometry (Fig. 3). EG&G Ortec model 676A 'Alpha King' α spectrometer with a Canberra 600 mm² passivated ion-implanted planar silicon (PIPS) detector was used. The spectrometer was connected to an EG&G Ortec model 476-4 Multiplexer, EG&G Ortec model 917 ADCAM Multichannel Buffer, EG&G model 918-DFP Dual Port Fanout and an IBM AT personal computer for multichannel analyzer emulation.

Samples were placed close to the detector and counted for 200–1000 min. Detection efficiency was $0.321 \pm 3\%$. Typical sample resolution (FWHM) was 30 keV suitable for analyzing both α -emitting radionuclides without significant peak overlapping.

Analysis of ^{210}Pb by LSC (Fig. 4). A Packard-Canberra model TRI-CARB 2250CA liquid scintillation counter was used in low-level count mode. Samples were counted in three regions (A: 0–18, B: 18–50, C: 50–150 channels) corresponding to the radiations of ^{210}Pb + low energy region of ^{210}Bi , to the middle energy region of ^{210}Bi and to the high energy region of ^{210}Bi + ^{210}Po , respectively. Samples and background

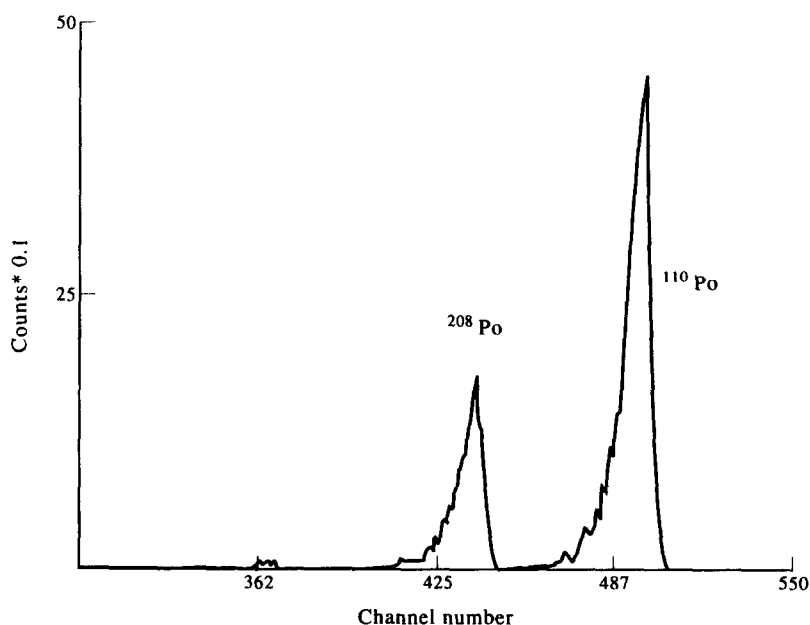


Fig. 3. Alpha spectrum of polonium source prepared from 5.0 g of IAEA-308 reference material (seaweed). Live time = 67901.2 s; data collected at 13:50 on 9 June 1994.

were counted for 200–1000 min. To determine the background, an inactive sample containing lead-oxalate, 1 ml 6 M HNO₃ and 14 ml of Insta gel was prepared. Measured count rates were always corrected for the “background” count rate determined in this manner. (It was also checked that lead-oxalate had no quenching effect.)

The LSC instrument was calibrated using a standard (equilibrium) ²¹⁰Pb–²¹⁰Bi–²¹⁰Po solution of known activities. Bismuth-210 was separated from an aliquot of the standard solution using the Sr. Spec crown ether, and a pure ²¹⁰Pb source was prepared with the same composition as ²¹⁰Pb sources and counted with the LSC. The ratio of the count rates in region A to region B was determined. This factor (0.684) was used to correct the count rates measured in region A to determine the contribution of ²¹⁰Pb to the total count rate (without the ²¹⁰Bi growing into the ²¹⁰Pb). Similarly, a correction factor for the determination of pure ²¹⁰Po in region C can be derived. Counting efficiency for ²¹⁰Pb in region A was determined with a ²¹⁰Pb(–²¹⁰Bi) source prepared from an aliquot of the standard solution. 30 mg of Pb carrier spiked with a known amount of ²¹⁰Pb tracer were separated with Sr. Spec crown ether using the same method as described above. A lead-oxalate source was prepared and the recovery (*r*) was determined by gravimetry. Thus the activity of ²¹⁰Pb

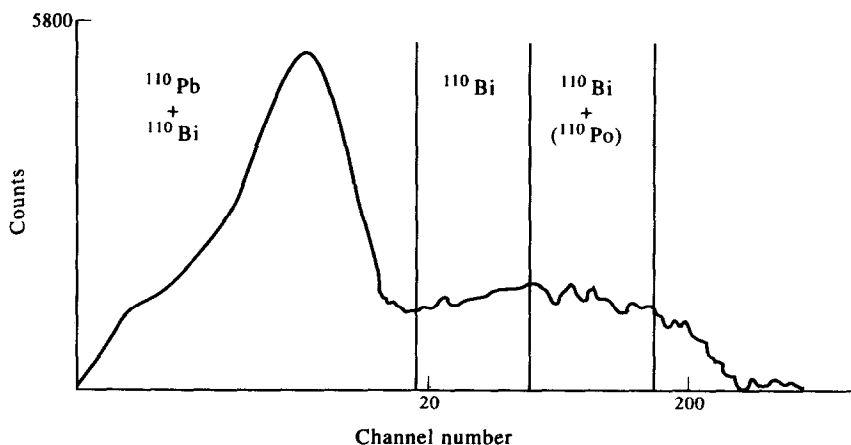


Fig. 4. LSC spectrum of lead source prepared from 5.0 g of IAEA-308 reference material (seaweed).

Low level Count Mode

	LL	UL	Count rate [cpm]	Uncertainty 2σ [%]
Region A	0.0	18.0	20.3	2.46
Region B	18.0	50.0	4.2	5.44
Region C	50.0	150.0	4.5	5.25

Quench parameter: tSIE = 148 (ES terminator = 60 s)

Live time = 400 min. Data printed at: 10:09 on 10 June 1994

was calculated and the counting efficiency for ^{210}Pb in region A was obtained. According to our measurements the counting efficiency was 0.759. Activity concentrations of both ^{210}Pb and ^{210}Po calculated in Bq/kg dry weight were corrected for chemical recoveries and cooling times.

ICP-AES measurements. Column effluent and strip solutions were analyzed by inductively coupled plasma atomic emission spectroscopy for some elements. A Jobin Yvon S38 instrument was used. The concentrations in the samples were related to those of the reference (lead) sample that was analyzed by the same method.

RESULTS AND DISCUSSION

Distribution ratios for lead, polonium and bismuth from HCl solutions have been determined by batch experiments. Measured distribution ratios plotted against HCl molarities are shown in Fig. 1. Although the results cannot be directly compared with the weight distribution ratios from nitric acid solutions obtained by Horwitz *et al.* because of the different experi-

mental conditions, it can be stated that lead is well retained by the Sr. Spec from moderately concentrated HCl solutions. The high distribution ratio (> 400) in solutions of 0.5–2 M HCl indicates a strong, but reduced Pb retention compared to HNO_3 at the same concentrations, thus allowing the easier stripping of lead from the column. In 6 M HCl, lead is not retained, so this solution can be used as a proper stripping agent of lead. The better polonium is retained, the higher the HCl concentration is. From solutions of more than 2 M HCl the distribution ratio is above 100 allowing its separation from bismuth and possibly other non-retained elements. Polonium can be easily stripped from the column with 6 M HNO_3 . It is worth mentioning that, in principle—regarding the non-selective LSC detection of ^{210}Pb and the highly selective alpha spectrometric detection of ^{210}Po —a highly selective separation procedure is needed for lead, while a less selective one is suitable for polonium. This separation demand seems to be fulfilled by the Sr. Spec in HCl.

Separation properties of lead, bismuth and polonium have been studied by column experiments. A typical elution curve obtained by the separation of a ^{210}Pb – ^{210}Bi – ^{210}Po tracer solution containing 15 mg of lead carrier is shown in Fig. 2. Bismuth-210 is not retained by the Sr. Spec from 2 M HCl; it is found in the effluent and the first 30 ml of eluate. The next 70 ml of 2 M HCl eluate are free of radioactivity. Polonium can be stripped with 60–70 ml of 6 M HNO_3 , while lead is separately stripped with 60 ml of 6 M HCl.

The chromatographic behaviour of several other elements has been studied by the separation of the standard reference material, lake sediment NBS 4354. 10 ml of 2 M HCl solution originating from the dissolution of about 0.5 g of dry sample, containing about 15 mg lead carrier and 4 dpm ^{208}Po tracer were loaded onto the column and treated in the same way as in the tracer experiment. Collected fractions were analyzed by ICP-AES. The elemental compositions of the fractions are shown in Table 1. Alkali metals are not retained by the crown ether. Other matrix components of soils and sediments, e.g. alkali earth metals (Mg, Ca), the earth metal Al, transition metals (Cr, Fe, Co, Cu, Zn, Ti, Zr, Y) and the non-metal P are not retained to a significant extent by the column; these elements behave similarly to bismuth. These components are removed by eluting the column with 20–30 ml of 2 M HCl, though a small part of them ($< 0.1\%$ of the load) can appear in the first strip solution, in the polonium fraction. However its presence usually does not interfere with polonium source preparation and ^{210}Po analysis. The lead fractions appear to be clear. In addition to the chromatographic separation, lead is finally purified by the lead-oxalate precipitation prior to gravimetric yield determination and LSC. Among the elements studied, only strontium and barium show a

slight retention but both are removed with about 50 ml 2 M HCl. The ICP measurements confirmed the conclusions of the test experiments; lead was quantitatively recovered from the column with about 50 ml of 6 M HCl. Aliquots of the 6 M HNO_3 strip solutions were analyzed for polonium, also. Results confirmed the preliminary conclusions: polonium is stripped by 75 ml of 6 M HCl, and about 80% of it appeared in the second 25 ml.

Unfortunately, the behaviour of only a limited number of elements of the periodic table has been studied in detail. Further examinations are needed, especially for actinides which show a significant retention on the crown ether from nitric acid, etc.

To obtain indirect information about other elements and to test the total analytical procedure, reference materials of known ^{210}Pb and ^{210}Po concentrations and of different chemical compositions have been analyzed. Samples under study included different sediments, phosphate ore, seaweed and cockle flesh. It must be noted that there is a lack of certified reference materials for ^{210}Pb and ^{210}Po (the only exception was the IAEA's sediment 368), so materials of recommended or information values have also been analyzed.

Samples and the results of analyses are listed in Table 3. Depending on the expected concentration, sample amounts varied between 0.4 g and 35 g. It is recommended that samples should be dissolved directly, because during ashing at higher temperatures, lead and polonium losses may arise. According to our experiences that are far from being sufficient, losses of lead during ashing at 600°C are more likely for biological materials of high organic content than for soil or sediment samples, the latter being often below detection limit. (Compare the ^{210}Pb results of sediment 4354 and cockle flesh 134.) Losses of polonium itself during ashing have not been studied, because so-called 'old' ash samples (older than half a year) were analyzed where ^{210}Po already changed to ^{210}Pb .

Samples were digested according to the wet chemical procedure described in the experimental part. Losses especially for polonium in biological materials could not be excluded, although nitric acid was added to most of the destruction steps to keep polonium permanently in an oxidized state. The low recovery for polonium in the cockle flesh was likely due to the non-sufficient destruction conditions. Some experiments have been performed to combine the fast destruction method of pyrosulfate fusion according to Sill and co-workers with the crown ether separation. Preliminary results for small sample amounts are promising.

Lead and polonium were separated on the preconditioned Sr. Spec column according to the detailed procedure (see Table 2). The separation, including loading, elution and stripping, is fast; it can be carried out within 2.5 h (the flow rate is > 2 ml/min). The volume of the load solution

TABLE 3
Activity Concentrations of ^{210}Pb and ^{210}Po in Reference Materials

Reference material	Sample size [g]	^{210}Pb		^{210}Po		Weighed average and uncertainty of ^{210}Pb and ^{210}Po concentrations (Bq/kg dry)		^{210}Pb	
		Recovery [%]	Activity concentration and uncertainty (Bq/kg dry)	Recovery [%]	Activity concentration and uncertainty (Bq/kg dry)			Reference value (Bq/kg dry)	Confidence interval (Bq/kg dry)
IAEA-308 Sediment	10.0	67	20.3 $\pm$ 1.8	67	21.1 $\pm$ 1.1	20.9 $\pm$ 0.94	23.2	19.8–27.2	Reference Information
NIST-4354 Sediment	0.43 aged ash	50	141.7 $\pm$ 27.7	64	138.8 $\pm$ 6.8	139.0 $\pm$ 6.6	120		
IAEA-135 Sediment	5.0	70	71.1 $\pm$ 5.50	55	71.95 $\pm$ 4.10	71.64 $\pm$ 3.28	48	42.2–54.13	Information
IAEA-308 Sediment	5.0	70	64.80 $\pm$ 4.99	58	69.34 $\pm$ 3.95	67.59 $\pm$ 3.10			
IAEA-308 Sediment	5.0	73	76.4 $\pm$ 5.9	70	85.4 $\pm$ 4.9	81.7 $\pm$ 3.8	73.0	66.0–75.0	Recommend Information
IAEA-134 Cattle flesh	35	76	3.46 $\pm$ 0.77	20	4.02 $\pm$ 0.23	3.99 $\pm$ 0.23	5.4	3.3–7.0	Information
Phosphate Ore	3.67 aged ash	57	2.35 $\pm$ 0.55	54	2.61 $\pm$ 0.14	2.59 $\pm$ 0.14	1292		
	1.0	87	1464 $\pm$ 87	45	1477 $\pm$ 82	1471 $\pm$ 60			
	1.0	97	1486 $\pm$ 88						
	1.0	92	1477 $\pm$ 87						
	1.0	91	1460 $\pm$ 86						
	1.0	97	1577 $\pm$ 93						
Average			1493 $\pm$ 48						

should not be less than 20 ml, otherwise lead precipitates easily. For stripping polonium, 6 M HNO_3 is used. The first 25 ml are rejected because polonium is stripped only with the further 60 ml of acid. Lead is well stripped with 60 ml 6 M HCl .

Sr. Spec columns can be reused after a special treatment: they are washed with 50 ml distilled water, 100 ml 0.1 M EDTA-ammonia (pH = 7) solution, 50 ml of distilled water and 100 ml of 2 M HCl . Care must be taken during column regeneration, because the column capacity is gradually reduced probably due to the destruction of the crown ether molecule by concentrated nitric acid, the Sr. Spec material is light sensitive and cross-contamination in extreme cases can appear. With special care columns can be reused. (We have good experience with up to 10 experimental runs with one column.)

Polonium sources are prepared by the selective method of spontaneous deposition from 0.1 M HCl onto silver discs. The procedure is well-known from the literature. As a result of the chromatographic separation, there are almost no interfering components during deposition, and with careful pH adjustment utilizing NaOH , the polonium plates show good resolution in alpha spectrometry (<30 keV) and the recoveries are also high. According to test experiments, recoveries are >85% when the deposition is carried out at 80°C for 90 min. Small amounts of iron or calcium do not disturb the deposition. Of course, the overall polonium recoveries for the procedure are smaller, typically between 50 and 70%.

Lead is prepared as lead oxalate; it has a low solubility and a well-defined stoichiometric composition for gravimetric yield determination. It is colourless, having no quenching effect in LSC, and the precipitation represents a further separation step of lead. Although a simple evaporation of the lead strip solution would provide somewhat higher recoveries for lead as chloride, because of the above-mentioned reasons, lead oxalate is preferred. Typical lead recoveries from real samples as lead oxalate are 60–80%. (Recoveries higher than 90% were obtained only by lead chloride precipitation.) Lead samples were counted by LSC. To have the same quench, lead oxalate was always dissolved with 1 ml 6 M HNO_3 and mixed with 15 ml Insta gel. (Quench parameter was checked during counting.) Counting of the samples was started 2 days after lead separation so that the longest lived lead radionuclide of the natural radioactive series, the ^{212}Pb (half-life 10 h) should basically decay. Counting times were selected up to 1000 min. During this time interval ^{210}Bi grows into ^{210}Pb to a significant extent. However, from the total count rate in the common region of ^{210}Pb and ^{210}Bi (region A), the contribution of ^{210}Bi is subtracted (see Fig. 3). Bismuth-210 is also counted in region B which does not contain other nuclides. The count rate of ^{210}Bi in region A can be calcu-

lated, because the shape of the beta spectrum is constant (see the experimental section). Applying these measuring and calculation methods, counting times are increased, thus reducing counting errors and the detection limit for ^{210}Pb . A detection limit of about 1 Bq/kg can be reached in the case of 10 g of sample of using a commercial LSC.

In Table 2 activity concentrations are given in Bq/kg calculated for the reference time and for dry weight. Uncertainties were calculated according to the law of error propagation and they include counting errors, calibration errors, weighing errors, etc. The major error components for most cases are counting error and calibration error for ^{210}Pb and ^{210}Po determinations, respectively. Typical uncertainties of ^{210}Pb and ^{210}Po analysis are 7–8% and 5–6%, respectively. The somewhat smaller error of the latter originates from its higher sensitivity due to the lower background of alpha spectrometry. For all the samples under study, a radioactive equilibrium between ^{210}Pb and ^{210}Po can be estimated; the common activity concentration was calculated, also, as a weighted average with a weighted uncertainty. The weighted uncertainties are smaller than any of the original values; typically, they are less than 5%. These final results were compared with the reference or information values.

Activity concentrations of ^{210}Pb and ^{210}Po listed in the same row of Table 3 originate from the same sample, with the same destruction and the common separation procedure. Sources are prepared and analyzed separately, by LSC and alpha spectrometry, respectively. Calibration of the two measuring systems was independent. Both methods have their own ways of chemical yield determinations (gravimetry prior to LSC and isotope dilution for alpha spectrometry). According to these points of views, the agreement of the ^{210}Pb and ^{210}Po concentrations can be regarded as good; for each material, it is within the measuring uncertainty.

The measured activity concentration of ^{210}Pb - ^{210}Po in the standard reference material, sediment IAEA-368 agrees well with the reference value. The agreement for the reference materials that do not have certified ^{210}Pb concentrations is less satisfying: the difference varies between 10% (seaweed or phosphate ore) and 80% (sediment IAEA-135) both in positive and negative directions excluding a systematic error. A set of five measurements was carried out to determine the reproducibility of the method: lead concentration of the phosphate ore which was tested in an intercomparison in the Netherlands gave the result of (1493 ± 48) Bq/kg, i.e. a reproducibility of about 3%. The measured value agrees well with the result of the intercomparison run (1292 Bq/kg), also. (Reproducibility of the analysis of the sediment IAEA-135 is also within the measuring error.)

CONCLUSION

Regarding the good reproducibilities, the agreement of the measured ^{210}Pb and ^{210}Po concentrations with each other and with the certified reference value, the method should be used for the accurate and sensitive determination of both nuclides in environmental and biological samples.

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REFERENCES

- Alfassi, Z. B. & Wai, C. M. (1992). *Preconcentration Techniques for Trace Elements*, CRC Press, Boston, MA.
- Barnabee, R. P. & Sill, C. W. (1981). Radiochemical Determination of Polonium-210 in Ores and Environmental Samples, Report No. IDO-12092, Radiological and Environmental Sciences Laboratory, Idaho Falls.
- Church, T. M., Hussain, N., Ferdelman, T. G. & Fowler, S. W. (1994). An Efficient Quantitative Technique for the Simultaneous Analysis of Radon Daughters ^{210}Pb , ^{210}Bi and ^{210}Po , *Talanta*, **41**(2), 243–249.
- HASL Procedures Manual, Health and Safety Laboratory, U.S. Atomic Energy Commission.
- Horwitz, E. P., Chiariza, R., Dietz, M. L. (1992). *Solvent Extr. Ion Exch.*, **10**, 313.
- Horwitz, E. P., Dietz, M. L. & Fischer, D. E. (1991). *Anal. Chem.*, **63**, 522.
- Horwitz, E. P., Dietz, M. L., Rhods, S., Felinto, C., Gale, N. H. & Houghton, J. *Anal. Chim. Acta*, **292**, 263.
- Moser, R. N. (1993). A Comparison of Methods for the Determination of the Dating-Nuclides ^{210}Pb and ^{226}Ra . *J. Radioanal. Nucl. Chem.*, **173**(2), 283–292.
- RADREM, (1989). Sampling and Measurement of Radionuclides in the Environment, Report by the Methodology Sub-group to the Radioactivity Research and Environmental Monitoring Committee, HMSO, London.
- Sebasta, F., Braun, T. & Ghersini, G. (1975). *Extraction Chromatography*, Elsevier, New York, Chapter 11.
- Steinhausser, F. (1990). Technologically Enhanced Natural Radiation and the Significance of the Related Risks, Proc. of Int. Conference on "High Levels of Natural Radiation", Ramsar, 3–7 Nov 1990, pp. 163–176.
- U.N. (1988). Report; Ionizing Radiation: Sources and Biological Effects, UNSCEAR, United Nations, New York.

- USEPA (1979). Radiochemical Analytical Procedures for Analysis of Environmental Samples, Report No. EMSL-LV-0539-17, U.S. Environmental Protection Agency, Environmental Monitoring and Support Laboratory, Las Vegas.
- Walton, H. F. & Rocklin, R. D. (1990). *Ion Exchange in Analytical Chemistry*, CRC Press, Boston, MA, Chapter 9.
- Yamamoto, M., Abe, T., Kuwabara, J., Komura, K., Ueno, K. & Takizawa, Y. (1994). *J. Radioanal. Nucl. Chem.*, **178**(1), 81.